Nicolò Santilio

Automation & IoT Engineer | Full-Stack Developer | Edge-to-Cloud, ROS2, Quarkus, Kubernetes

Bari, Italy nicolo.santilio@outlook.com +39 346 332 1097 nicolosantilio.it nicolosantilio fenix-hub

Professional profile

Automation and IoT Engineer with a full-stack background, experienced in designing *edge-to-cloud* and *digital twin* architectures for real-time control systems. Strong hands-on experience with ROS2, Quarkus, containerization, and modern DevOps pipelines. Passionate about robotics, distributed systems, and secure IoT integration.

Technical skills

IoT & Automation: ROS2, Gazebo, Digital Twin, MQTT, OPC-UA, Modbus/TCP, Edge Computing, RTOS, Arduino, Raspberry

Languages: Java, TypeScript/JavaScript, Golang, Lua, GDScript, Python, C++, SQL, NoSQL, HTML, GraphQL, Matlab

Frameworks: Quarkus, Spring Boot, Angular, Node.js, Vue.js, Vaadin, Express.js, Tailwind, Flask, Gin

Integration tools: Camunda, Zeebe, WSO2, Kong Gateway, KrakenD, Tyk, Traefik, HA-Proxy, Nginx, Caddy

Cloud, DevOps & CI/CD: Git, Docker, Kubernetes, GitHub Actions, Vercel, Render, DigitalOcean, Observability

Communication patterns: RESTful, gRPC, WebSocket, Server-Sent Events

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Ethereum Blockchain, Elasticsearch, OpenSearch

Platforms: Firebase, Supabase, Appwrite

Data & AI: RLlib, Model Identification, Time-series Analysis, LangChain, Ollama, vLLM, llama.cpp, LiteLLM, Langfuse, Open WebUI

Security, protocols & standards: OIDC-FED (OpenID Connect Federation 1.0), OpenID.Core, RFC7519 (JWT), RFC7523 (JWT OAuth Profile), RFC7636 (PKCE), RFC7638 (JWK Thumbprint), RFC7662 (OAuth Token Introspection), RFC9068 (JWT OAuth Access Token Profile), EN319-412-1 (Electronic signatures), Real-time Streaming Protocol (RFC7826), MQTT (RFC9431), W3C WebAuthn, FIDO CTAP2

Work experience

Solution Architect *Jan 2024 – Present*
Auriga SpA, Bari (Italy)

- **Architecture of an on-premise infrastructure for the execution and monitoring of workflows and agent-based applications, aimed at enhancing the software development process and automating internal processes supported by generative AI** – Cloud-native Quarkus microservices in a Docker environment, Langfuse, vLLM, Ollama, LiteLLM, LangChain, n8n. .
- **Orchestration of legacy and modern services with Quarkus and BPM tools** – design of secure APIs and workflows across heterogeneous systems. *84.4% execution-time reduction and 91.3% error-rate reduction, achieved by integrating and orchestrating legacy systems and heterogeneous resources within the "AURIGA-Ecological Digital Conversion - AURIGA-EDICO" project (4WBHY73 - AURIGA EDICO).*
- **Development of a white-label SaaS platform for selling and delivering digital services** – designed to support an ecosystem of companies and developers with a turnkey "ready-to-use" configuration, within the AURIGA PBM project (PROJECT CODE 98VZ3X5 – AURIGA-PBM). *Quarkus, Docker, Kubernetes, Angular, Go.*
- **Delivery of internal training on Business Process Management** – technical and managerial audiences. *improved adoption and delivery speed.*
- **Development of microservices in the PagoPA ecosystem to deliver new features** – cloud-native Quarkus microservices on Kubernetes, AWS, GitHub CI/CD, Terraform. .

R&D Engineer (NDA) *May 2022 – Jan 2023*
W4 Games Limited, Remote

- **Design and implementation of *serverless* backends** – Supabase, SQL. *optimized for real-time performance and low latency.*

Software Engineer & Co-Architect (Apprenticeship) *Jan 2021 – Dec 2023*
Auriga SpA, Bari (Italy)

- **Development of an e-commerce platform for digital services** – Industry 4.0, microservices architecture. *improved efficiency of production processes.*
- **Definition of the microservices stack and domain models** – SAGA, CQRS, API composition. *increased modularity and fault tolerance.*
- **Contractual certification on blockchain** – Hyperledger Sawtooth. *secure transactions between suppliers and consumers.*

Education

1st-level Master | Industrial IoT *May 2026 – Jul 2026*
Polytechnic University of Bari

- **Coursework:** Intensive program covering IIoT communication technologies (MQTT, OPC-UA, wide-area and short-range protocols), microservice orchestration, edge/cloud computing, and AI-driven control systems. Hands-on focus on industrial digital twin architectures, SaaS/PaaS development, and real-time process automation.(Program details)

M.Sc. Automation Engineering | Cyber Physical Systems *Sep 2021 – Jul 2026*
Polytechnic University of Bari

- **Coursework:** Optimization and Control; Advanced Software Engineering; Secure Programming Lab; Artificial Intelligence for Automation; Networked Control Systems; Model Identification and Intelligent Data Control; Final Project; Internship; Model Predictive Control (MPC); Robotics; Artificial Intelligence and *Machine Learning*; Big Data; *Business Process Management*; Distributed Measurement and Data Acquisition Systems; Dynamical Systems Theory; Embedded Control; Estimation and Control of Dynamical Systems; Internet of Things. (Program details)

1st-level Master | Sustainable and Digital Process Transformation *Feb 2025 – Apr 2025*
Polytechnic University of Bari

- **Coursework:** Methods for *Business Process Management*; Sustainable process transformation; Digital process transformation; Project work and final exam. (Program details)

B.Sc. Computer and Automation Engineering *Sep 2016 – Jun 2020*
Polytechnic University of Bari

- **Coursework:** Algorithms and Data Structures in Java; Mathematical Analysis; Numerical Computing; Business Economics and Organization; Physics 1–3; Geometry and Algebra; Computing for Engineering; Databases and Information Systems; Electrical Engineering; Fundamentals of Automatic Control; Fundamentals of Electronics and Electronic Devices; Operating Systems; Optimization Methods; Industrial Automation; Electrical Machines; Applied Mechanics; Digital Control; Measurements; Fundamentals and Telecommunications Networks. (Program details)
- **Thesis:** *Knowledge Representation and Automated Reasoning for Affective Computing from Biosignals.* (PDF)

Selected projects

Digital Twin for an Ackermann Robot (ROS2/Nav2) *mobile-robotics*

- **Implemented an autonomous navigation stack with Gazebo simulation** – ROS2 Nav2, OSM map. *fast and repeatable scenario iteration.*

Open Authenticator (HOTP/TOTP) *open-authenticator*

- **Developed OTP libraries *from scratch* compliant with standards** – Java, C++. *cross-platform, privacy-oriented 2FA.*

Wearable system for VOC monitoring *firefliot*

- **Built a wearable industrial environmental monitoring IoT system** – MQTT, Arduino, Raspberry Pi. *real-time data acquisition.*

QRLog — QR-based authentication *qr-log-demo*

- **Implemented QR login with real-time communication** – Quarkus, WebSocket, DDD. *secure login flow in real time.*

SRAP — Transradial prosthesis control *srap*

- **Developed a control algorithm for an EMG-based prosthesis** – C, STM32 Nucleo. *hardware integration and neurophysiological analysis.*

CNS Live — Real-time streaming platform *cns-live*

- **Built an open-source platform for live streaming and VOD with RTMP and HLS/DASH** – Angular, Node.js, ffmpeg, WebSockets. *real-time user metrics and adaptive streaming.*

BariNAV — RL agent for route planning *ai-for-automation*

- **Designed a *reinforcement learning* agent for optimal routes in urban environments** – Python, RLlib, SUMO, TomTom API. *integration of traffic data for realistic simulations.*

Apulia4All — Accessibility web app (Hackathon) *apulia-4-all*

- **Developed an inclusive web app integrating environmental and accessibility datasets** – Svelte, Node.js, TypeScript, OpenStreetMap. *presented at OpenTusk 2023.*

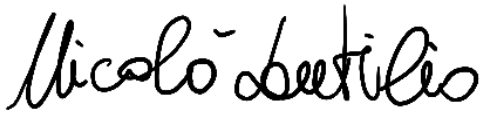
GDCache — Caching library for GDScript *gdcache*

- **Created a caching library with replacement policies and performance metrics** – GDScript. *open-source project with abstract extension points.*

Publications

- **Development of a digital twin for simulating autonomous navigation of an Ackermann-steering vehicle** *(Jun 2025)*
- **Development of a multi-objective reinforcement-learning (RL) agent for route planning in urban environments** *(Dec 2024)*
- **Development of an interactive web platform for livestreaming based on ffmpeg and adaptive video format** *(Dec 2023)*
- **Design of a black-box analysis and identification model of a permanent-magnet synchronous motor in steady-state operation** *(Jul 2023)*
- **Prototyping and control of a transradial robotic prosthesis based on surface electromyography** *(Oct 2022)*
- **Wearable IoT system for personalized VOC monitoring in manufacturing companies** *(Jul 2021)*

- **Knowledge representation and automated reasoning for affective computing from biosignals** *(Jun 2019)*

I hereby authorize the processing of my personal data contained in this CV pursuant to Italian Legislative Decree No. 196 of 30 June 2003 and the GDPR (Regulation (EU) 2016/679).

Bari, 12 February 2026